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Logistic Regression

LR is a binary classification algo that predicts the prob. of a class (mostly binary 0/1).

Ex: Spam / Not spam.

$$y \in (0,1).$$

$$\hat{y} = P(y=1/x).$$

For k features $x_1, x_2, x_3, x_4, \dots, x_k$.

$$x_i = x_{i1} + x_{i2} + x_{i3} + \dots + x_{ik}.$$

$$w = [w_1, w_2, w_3, \dots, w_k].$$

$$z_i = w_1 x_{i1} + w_2 x_{i2} + \dots + w_k x_{ik} + b$$

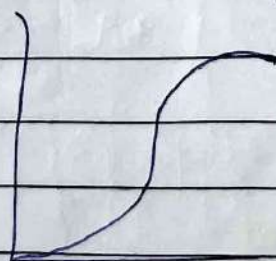
$$\hat{y}_i = \sigma(z_i) = \frac{1}{1 + e^{-(z_i)}}$$

$$y_i = \begin{cases} 1 & \text{if } \hat{y}_i \geq 0.5 \\ 0 & \text{if } \hat{y}_i < 0.5 \end{cases} \quad \text{Decision Rule}$$

Loss Fxn : (Binary Cross Entropy loss)

$$J(w, b) = -\frac{1}{n} \sum_{i=1}^n [y_i \log \hat{y}_i + (1-y_i) \log (1-\hat{y}_i)]$$

PTO →



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Gradient wrt w_k :

$$\frac{\partial J}{\partial w_j^0} = \frac{1}{m} \sum_{i=1}^m (\hat{y}_i^0 - y_i^0) x_{ij}^0$$

Gradient wrt to Bias:

$$\frac{\partial J}{\partial b} = \frac{1}{m} \sum_{i=1}^m (\hat{y}_i^0 - y_i^0)$$

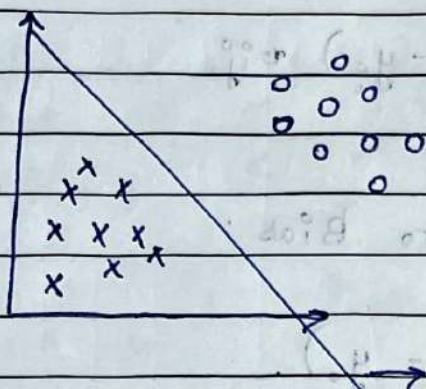
Update:

$$w_j^0 = w_j^0 - \alpha \frac{\partial J}{\partial w_j^0}$$

$$b = b - \alpha \frac{\partial J}{\partial b}$$

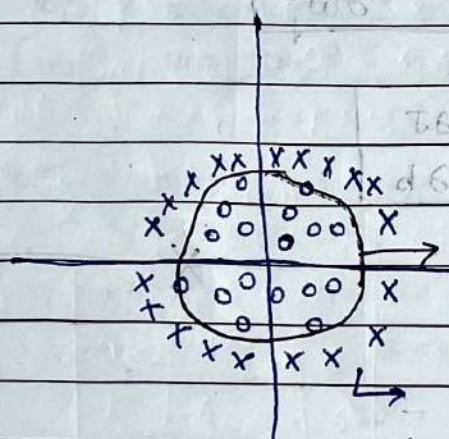
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Decision Boundary



Decision Boundary
is linear.

If Data is linearly ~~separable~~ separable
we use the LR model
→ linear Regression



Decision
Boundary.

Not linearly separable
Data.

$$f = w_1 x_1^2 + w_2 x_2^2 + b.$$

Let $w_1 = 1$, $w_2 = 1$, $b = -2 \rightarrow$ After
train.

$$g(w_1 x_1^2 + w_2 x_2^2 + b).$$

$$f = w_1 x_1^2 + w_2 x_2^2 + b.$$

$$x_1^2 + x_2^2 - 2 \rightarrow \text{Put } f = 0.$$

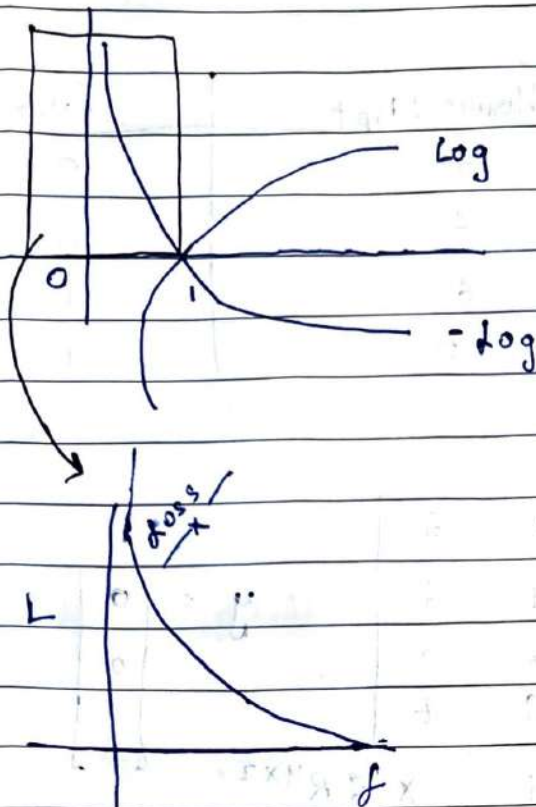
$$x_1^2 + x_2^2 = 2$$

Circle

Eq. of the Decision
Boundary

Log Graph

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Code : logistic Reg from Stattech.
Mathematics of Logistic Reg.
TP / TN / FP / FN

Confusion Matrix

- Precision
- Recall
- F1 Score
- Accuracy

Ex: Dataset

Hours Studied	Hours Slept	Pass
2	5	0
4	6	0
6	6	1
8	7	1

We Define :

$$X = \begin{bmatrix} 2 & 5 \\ 4 & 6 \\ 6 & 6 \\ 8 & 7 \end{bmatrix}$$

$$y = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 1 \end{bmatrix}$$

No of Samples = 4 $X \in \mathbb{R}^{4 \times 2}$

No of features = 2 $y \in \mathbb{R}^{4 \times 1}$

Add Bias Column

$$X = \begin{bmatrix} 1 & 2 & 5 \\ 1 & 4 & 6 \\ 1 & 6 & 6 \\ 1 & 8 & 7 \end{bmatrix}$$

Shape becomes $X \in \mathbb{R}^{4 \times 3}$

Each row becomes $x_i = [1, x_{i1}, x_{i2}]$

Eg,

$$x_1 = [1, 2, 5]$$

$$x_2 = [1, 4, 6]$$

Initialize weight

$$w = \begin{bmatrix} b \\ w_1 \\ w_2 \end{bmatrix}$$

$$w = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Linear Combination:

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Compute:

$$z = Xw$$

$$z = \begin{bmatrix} 1 & 2 & 5 \\ 1 & 4 & 6 \\ 1 & 6 & 7 \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$z = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} \rightarrow w_i x_i + b$$

Each prediction before sigmoid

$$z_i = w^T x_i$$

Sigmoid $\hat{y} = \sigma(z)$

Since,

$$z = 0$$

$$\sigma(0) = 0.5$$

$$\hat{y} = \begin{bmatrix} 0.5 \\ 0.5 \\ 0.5 \\ 0.5 \end{bmatrix}$$

Loss $\hat{y}$

$$J(w) = -\frac{1}{m} \sum_{i=1}^m [y_i \log(\hat{y}_i) + (1-y_i) \log(1-\hat{y}_i)]$$

where $m=4$

$$y_i = 0 \quad \hat{y}_i = 0.5 \quad \text{for All points}$$

$$J = \frac{1}{4} (0.693 + 0.693 + 0.693 + 0.693)$$

$$J = 0.693$$

Gradient Computation.

$$\nabla J = \frac{1}{m} X^T (\hat{y} - y)$$

$$\hat{y} - y = \begin{bmatrix} 0.5 \\ 0.5 \\ 0.5 \\ 0.5 \end{bmatrix} - \begin{bmatrix} 0 \\ 0 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 0.5 \\ 0.5 \\ -0.5 \\ -0.5 \end{bmatrix}$$

$$X^T = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 2 & 4 & 6 & 8 \\ 5 & 6 & 6 & 7 \end{bmatrix}$$

$$X^T (\hat{y} - y)$$

$$\Rightarrow \begin{bmatrix} 1 & 1 & 1 & 1 \\ 2 & 4 & 6 & 8 \\ 5 & 6 & 6 & 7 \end{bmatrix} \begin{bmatrix} 0.5 \\ 0.5 \\ -0.5 \\ -0.5 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 1(0.5) + 1(0.5) + 1(-0.5) + 1(-0.5) \\ 2(0.5) + 4(0.5) + 6(-0.5) + 8(-0.5) \\ 5(0.5) + 6(0.5) + 6(-0.5) + 7(-0.5) \end{bmatrix}$$

$$\Rightarrow \nabla J = \begin{bmatrix} 0 \\ -4 \\ -1 \end{bmatrix} \Rightarrow \begin{bmatrix} 0 \\ -1 \\ -0.25 \end{bmatrix}$$

Update rule

 x

$$w = w - \alpha \nabla J$$

$$\alpha = 0.1$$

$$w = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} - 0.1 \begin{bmatrix} 0 \\ -1 \\ -0.25 \end{bmatrix}$$

$$w = \begin{bmatrix} 0 \\ 0.1 \\ 0.025 \end{bmatrix}$$

New w

Model

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Confusion Matrix

A confusion matrix is a table used to evaluate a classification model by comparing:

- Actual labels
- Predicted labels

Structure:

	Pred Positive	Predicted Negative
Actual Positive	TP	FN
Actual Negative	FP	TN

Accuracy

$$\text{Accuracy} = \frac{\text{Correct Prediction}}{\text{Total prediction}}$$

Mathematically

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

Precision

Out of the predicted positive how many were actually positive.

$$\text{Precision} = \frac{TP}{TP + FP}$$

Precision →

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Recall (Sensitivity)

Out of all actual positives, how many did the model detect?

$$\text{Recall} = \frac{TP}{TP + FN}$$

F1 Score :

F1 Score combines precision & recall into one metric.

It is the harmonic mean of precision & recall.

$$F1 = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$